

Smart Text Similarity Engine

2026-09-05 | nlp | intermediate

local://C:\Users\msoni029\DailyAIProjects\projects\ai-project-2026-09-05-smart-text-similarity

Description

A text similarity engine that combines TF-IDF, cosine similarity, and Jaccard distance to find semantically similar documents. Useful for plagiarism detection, document deduplication, and content recommendation.

Tech Stack

scikit-learn, pandas, nltk, numpy, flask

Features

- Multi-algorithm similarity scoring
- REST API
- Batch CSV processing
- Configurable thresholds

Sample Input

Two text strings or a CSV file with a 'text' column for batch comparison

```
python main.py --text1 "Machine learning is a subset of AI" --text2 "AI includes machine learning as a subfield"
python main.py --batch documents.csv --threshold 0.7
```

Sample Output

Similarity scores between 0.0 and 1.0 with method breakdown

```
Similarity Analysis Results:
Text 1: Machine learning is a subset of AI
Text 2: AI includes machine learning as a subfield

TF-IDF Cosine:  0.72
Jaccard Index:   0.45
Combined Score:  0.63
Verdict:        SIMILAR
```

Project Files

```
main.py
similarity_engine.py
requirements.txt
```

README.md
LICENSE
.gitignore
tests/test_engine.py

Test Results

PASSED

Run: 8 | Passed: 8 | Failed: 0

Pipeline

- [OK] Ideation (6.1s)
- [OK] Implementation (52.3s)
- [OK] Dependencies (8.7s)
- [OK] Test Generation (18.2s)
- [OK] Test Execution (1.2s)
- [OK] Report Generation (2.4s)
- [OK] Card Generation (1.1s)
- [OK] README Enrichment (0.3s)
- [OK] Portfolio Update (0.5s)
- [OK] Resume Update (0.4s)